```
# Upload the CSV file from your computer
from google.colab import files
uploaded = files.upload()
```

Choose Files Simulated_I…ored (1).csv

Simulated_IT-OT_Plant_Network_Risk_Assessment_scored (1).csv(text/csv) - 86244 bytes, last modified: 11/6/2026 - 100% done

```
Saving Simulated_IT-OT_Plant_Network_Risk_Assessment_scored (1).csv to Simulated_IT-OT_Plant_Network_Risk_Assessment_scored
```

```
# Load the file into a table and look at it
import pandas as pd

df = pd.read_csv("Simulated_IT-OT_Plant_Network_Risk_Assessment_scored (1).csv")
print(df.shape)
df.head()
```

```
(398, 20)
```

	finding_id	remediation_action	asset_type	segmentation_status	exposure_path	remediation_cost_inr	zone	assessm
0	ca63b253-ee30-4d1c-8757-10b1a548b8d5	Enforce MFA	Employee workstation	Isolated	NaN	45000	Enterprise IT	
1	ed8b8ca0-a3b1-4d14-9497-25c1e4b2c8d6	Quarterly patching	Employee workstation	Segmented	NaN	15000	Enterprise IT	
2	1904096e-11af-4a1f-9426-4ffa557ee558	Harden AD and MFA	Server	Partially segmented	NaN	180000	Enterprise IT	
3	52cdec6d-3f03-4a60-a1d2-6dcadf3bb028	Enable SIEM logs	Server	Segmented	NaN	90000	Enterprise IT	
4	5d12e609-12a9-4322-94ea-b52e355d026d	Restrict egress rules	Firewall	Partially segmented	NaN	120000	Enterprise IT	

Next steps: Generate code with `df` New interactive sheet

```
# Work out the four numbers for the top cards
total_findings = len(df)
critical_count = (df["risk_level"] == "Critical").sum()
open_paths = (df["segmentation_status"] == "Open path").sum()
total_cost_crore = df["remediation_cost_inr"].sum() / 10_000_000

print("Findings:", total_findings)
print("Critical:", critical_count)
print("Open paths:", open_paths)
print(f"Total fix cost: Rs {total_cost_crore:.2f} crore")
```

```
Findings: 398
Critical: 35
Open paths: 68
Total fix cost: Rs 5.40 crore
```

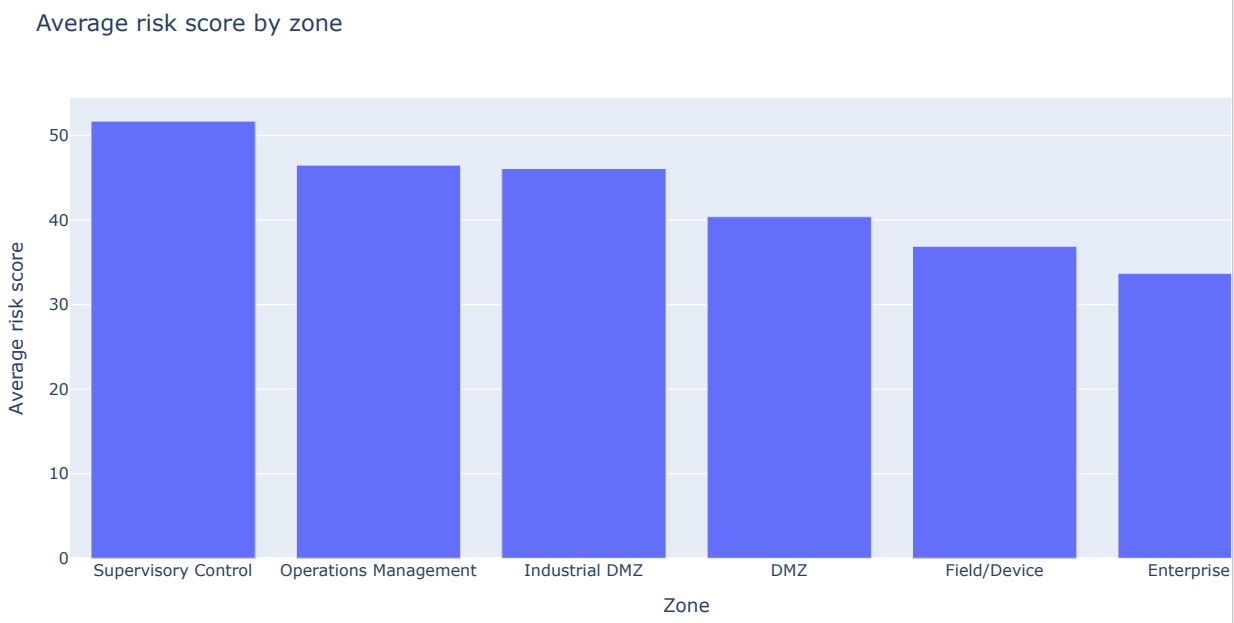
```
# Work out the average risk score for each zone
zone_risk = df.groupby("zone")["risk_score"].mean().round(1).sort_values(ascending=False)
print(zone_risk)
```

```
zone
Supervisory Control      51.7
Operations Management    46.5
Industrial DMZ           46.1
DMZ                      40.4
Field/Device             36.9
Enterprise IT            33.7
Name: risk_score, dtype: float64
```

```
# Draw the average risk by zone as a bar chart
import plotly.express as px

zone_df = zone_risk.reset_index()
fig = px.bar(zone_df, x="zone", y="risk_score",
             title="Average risk score by zone",
```

```
                labels={"risk_score": "Average risk score", "zone": "Zone"})
fig.show()
```




```
# Keep only the open path findings and count them by start zone and destination
open_df = df[df["segmentation_status"] == "Open path"]
paths = open_df.groupby(["exposure_path", "zone"]).size().reset_index(name="findings")
print(len(open_df))
paths
```

68

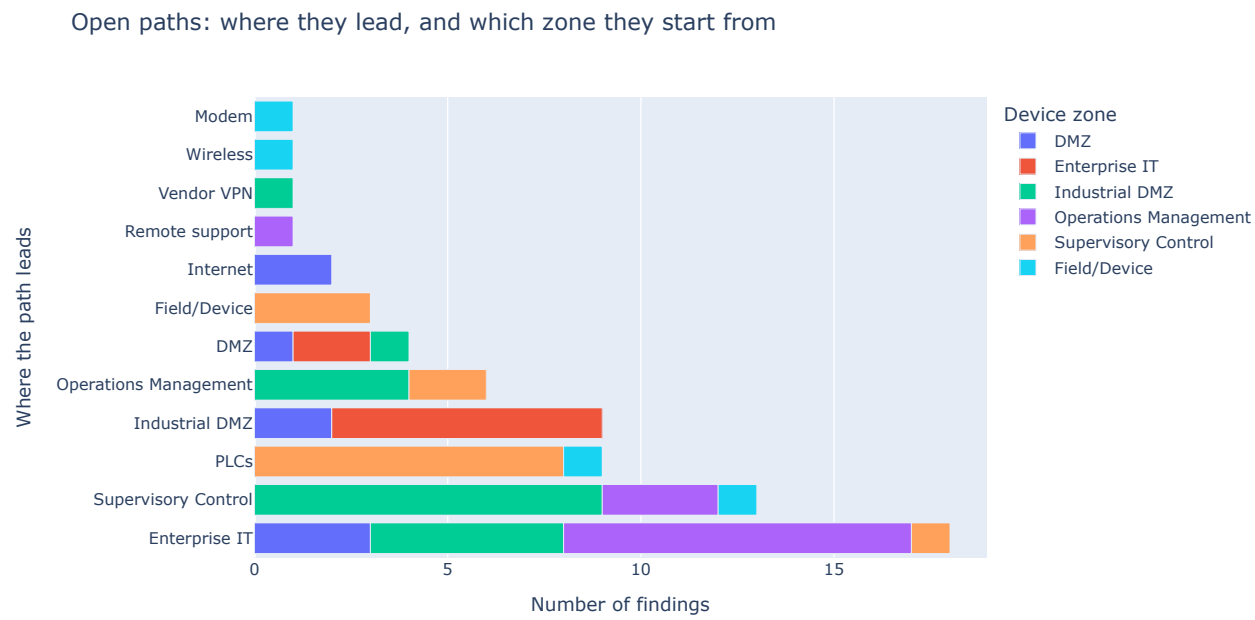
	exposure_path	zone	findings
0	DMZ	DMZ	1
1	DMZ	Enterprise IT	2
2	DMZ	Industrial DMZ	1
3	Enterprise IT	DMZ	3
4	Enterprise IT	Industrial DMZ	5
5	Enterprise IT	Operations Management	9
6	Enterprise IT	Supervisory Control	1
7	Field/Device	Supervisory Control	3
8	Industrial DMZ	DMZ	2
9	Industrial DMZ	Enterprise IT	7
10	Internet	DMZ	2
11	Modem	Field/Device	1
12	Operations Management	Industrial DMZ	4
13	Operations Management	Supervisory Control	2
14	PLCs	Field/Device	1
15	PLCs	Supervisory Control	8
16	Remote support	Operations Management	1
17	Supervisory Control	Field/Device	1
18	Supervisory Control	Industrial DMZ	9
19	Supervisory Control	Operations Management	3
20	Vendor VPN	Industrial DMZ	1
21	Wireless	Field/Device	1

Next steps: Generate code with paths | New interactive sheet

```
# Draw the open paths as a stacked horizontal bar chart
```

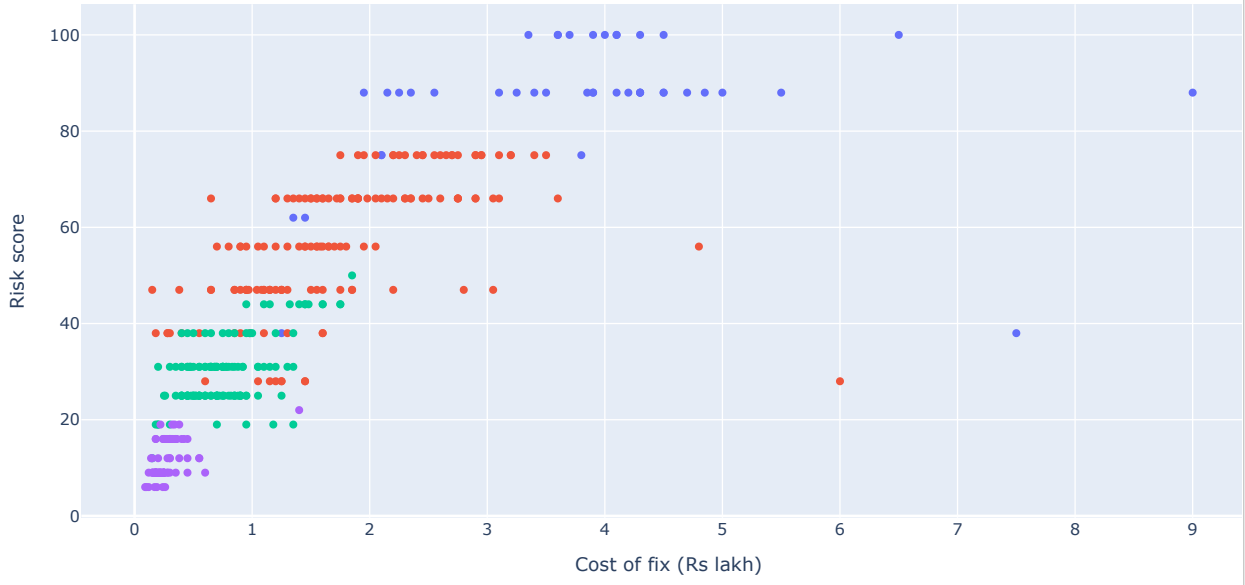
```
import plotly.express as px

order = paths.groupby("exposure_path")["findings"].sum().sort_values().index.tolist()
fig2 = px.bar(paths, x="findings", y="exposure_path", color="zone", orientation="h",
              category_orders={"exposure_path": order},
              title="Open paths: where they lead, and which zone they start from",
              labels={"exposure_path": "Where the path leads",
                      "findings": "Number of findings",
                      "zone": "Device zone"})
fig2.show()
```




```
# Add a cost column in lakh rupees and draw cost against risk
df["cost_lakh"] = df["remediation_cost_inr"] / 100_000

fig3 = px.scatter(df, x="cost_lakh", y="risk_score", color="operational_criticality",
                  hover_name="asset_name",
                  hover_data={"remediation_action": True, "zone": True, "cost_lakh": ":.1f"},
                  category_orders={"operational_criticality": ["Critical", "High", "Medium", "Low"]},
                  labels={"cost_lakh": "Cost of fix (Rs lakh)", "risk_score": "Risk score",
                          "operational_criticality": "Criticality"})
fig3.show()
```




```
# Rank the fixes: highest risk first, cheapest first when risk is tied
top10 = (df.sort_values(["risk_score", "remediation_cost_inr"], ascending=[False, True])
           .head(10)[["asset_name", "zone", "iec_62443_gap", "remediation_action",
```

```
                  "risk_score", "remediation_cost_inr"]])
top10
```

	asset_name	zone	iec_62443_gap	remediation_action	risk_score	remediation_cost_inr
325	MES-GW-02	Operations Management	SR 5.1 flat MES link	Separate MES interface	100	335000
238	CELL-SW-POWER-02	Field/Device	SR 5.2 route to PLCs	Deploy industrial firewall	100	360000
358	PROD-DB-01	Operations Management	SR 5.2 direct database link	Remove direct SQL link	100	360000
342	MES-DB-03	Operations Management	SR 5.1 direct corporate VLAN	Rebuild zone boundary	100	370000
308	MES-APP-04	Operations Management	SR 5.2 ERP direct access	Deploy one-way gateway	100	390000
374	MES-APP-06	Operations Management	SR 5.1 no IT-OT boundary	Insert OT firewall pair	100	400000
131	EWS-TURB-01	Supervisory Control	SR 5.1 dual-homed workstation	Remove dual-homed NIC	100	410000
225	RTU-YARD-01	Field/Device	SR 5.2 insecure wireless path	Replace wireless bridge	100	410000
397	SCADA-GW-04	Supervisory Control	SR 5.2 direct IT-to-PLC route	Enforce zone conduit controls	100	430000
232	PLC-TURBINE-	Field/Device	SR 5.1 shared	Redesign cell network	100	450000

Next steps: Generate code with top10 New interactive sheet